Today's agenda

- Problems
- for loops
- break / continue.

AlgoPrep

Q) Print Reverse

↳ Given an Integer N, Print all digits from Right to left.

ex: N = 3 7 2 6

↳ 6 2 7 3

//idea 1

N = 1 4 2 → lastdigit = N % 10 = 2

N = 2 6 3 4 8 → lastdigit = N % 10 = 8

N = 2 6 3 4 8

```
P S V main ( ) {
    Scanner scn = new Scanner(System.in);
        int n = scn.nextInt();
    if (n<0) { n = n*-1; }
        while (N>0){
            int lastdigit = N % 10;
            System.out.println (lastdigit);
              N = N/10;
        }
}
```

AlgoPrep

N = 26348

```
while (N>0){
    int lastdigit = N % 10;
        System.out.println (lastdigit);
          N = N/10;
}
```

N	lastdigit	
26348	~~8~~	8
2634	4	4
263	3	3
26	6	6
2	2	2
0		

↳ exit

5 times

↳ A variable created inside loop gets deleted once the iteration is complete.

AlgoPrep

// For loop basics

```
int i=0;
while (i<=10){
    //Statement

    i++;
}
```

```
for (int i=0; i<=10; i++){

    //Statements

}
```

initialization

condn

flow:

1 time: initialize i

Condition — false → exit the loop

true

execute the statement

update the variable

AlgoPrep

Q) Print numbers from 1 to 5 using for loop.

```
for (int i=1; i<=5; i++) {
        System.out.Println(i);
}
```

cond^n updation

i	i<=5
1	T
2	T
3	T
4	T
5	T
6	F

Limit

1 5
2
3
4

AlgoPrep

Q) Count factors

↳ Given a Positive number, Print all the factors of that number.

↳ Numbers Completely dividing N

Ex: N=12: 1 2 3 4 6 12

N=17: 1 17

N=24: 1 2 3 4 6 8 12 24

Minimum factor of N: 1

Maximum factor of N: N

//idea1

N=10

1 ⟶ N%1 ⇒ 10%1 = 0 ✓

2 ⟶ N%2 ⇒ 10%2 = 0 ✓

3 ⟶ N%3 ⇒ 10%3 = 1

4 ⟶ N%4 ⇒ 10%4 = 2

5 ⟶ N%5 ⇒ 10%5 = 0 ✓

6 ⟶ N%6 ⇒ 10%6 = 4

7 ⟶ N%7 ⇒ 10%7 = 3

8 ⟶ N%8 ⇒ 10%8 = 2

9 ⟶ N%9 ⇒ 10%9 = 1

10 ⟶ N%10 ⇒ 10%10 = 0 ✓

AlgoPrep

//Psuedo code

iteration count

N

```
P S V main () {
    Scanner scn = new Scanner (System.in);
    int n = scn.nextInt();

    for (int i=1; i<=N; i++) {
        if (N % i == 0) {
            System.out.println (i);
        }
    }
}
```

```
for (int i=1; i<=N; i++) {
    if (N % i == 0) {
        System.out.println (i);
    }
}
```

```
int i=1;
while (i<=N) {
    if (N % i == 0) {
        System.out.println (i);
    }
    i++;
}
```

Break till 9:38 PM

AlgoPrep

Q) IsPrime

↳ Given a number N, Prime "Prime" if the number is a Prime number else "Not Prime".

No. divisible by 1 & itself.

Ex: N = 3 → {1, 3} → Prime

N = 13 → {1, 13} → Prime

N = 30 → {1, 2, ... 30} → Not Prime

Quiz: N = 1 : Neither Prime nor Composite.

Idea

↳ Count the factors of N == 2 → Prime

else → Not Prime

N = 13 → Count of factor = 2 → Prime no.

N = 10 → Count of factor = 4 → Not Prime

AlgoPrep

//Psuedo code

```
P S V main () {
    Scanner scn = new Scanner (system.in);
    int n = scn.nextInt();
    int count=0;
    for(int i=1; i<=N; i++){
        if(N % i == 0){
            ~~System.out.println(i);~~ Count ++;
        }
    }

    if (Count == 2){
        System.out.println ("Prime");
    }
    else {
        System.out.println ("Not Prime");
    }
}
```

N iterations

AlgoPrep

// Break Statement

```
for (int i=1 ; i<=5 ; i++) {
    if (i==3){
        break;
    }
    System.out.println(i);
}
```

Not allowed to change

i
1
2
3
4
5

1
2
3

break: The moment you execute break statement you exit the current loop.

Ex:

```
for (int i=1; i<10; i++) {

    System.out.println(i);
    if (i==2){ break; }
}
```

i	i<10	
1	T	1
2	T	2

Quiz 3:

```
for (int i=0; i<5; i++){
        if (i>2) {break; }
        System.out.print (i+ " ");

}
```

i	i<5	if
0	t	f
1	t	f
2	t	f
3	t	t
		4 exit

→ 0 1 2

Quiz 4:

```
for (int i=0; i<5; i++){
        break;
        System.out.print (i+ " ");

}
```

i	i<5	
0	t	

AlgoPrep

//Continue Statement

↳ skip and go to next iteration.

Ex:

```
for (int i=0; i<5; i++){
        if (i==2){
            continue;
        }
        System.out.println(i);
}
```

i	i<5	if
0	t	f
1	t	f
2	t	t
3	t	f
4	t	f
5	f	

↳ exit

0 4
1
3

AlgoProp

Quiz 5:

```
for (int i=0; i<=5; i++){
        if (i==2 || i==3)
            Continue;
        3
        System.out.println(i);
}
```

i	i<=5	if
0	+	f
1	+	f
2	+	+
3	+	+
4	+	f
5	+	f
6	f	

↳ exit

0
1
4
5

AlgoPrep

Quiz 6:

&& → false dominated

```
for (int i=0; i<=5; i++){
        if (i==2 && i==3)
            Continue;
        3
        System.out.println(i);
}
```

i	i<=5	if
0	+	f
1	+	f
2	+	f
3	+	f
4	+	f
5	+	f
6	f	

↳ exit

0 4
1 5
2
3